

Working Memory: Effects of STSP on Stability and Flexibility in Attractor Networks

Sophia, Chris, Tireni, Geetika, Toka, Ziad, Yiyang, Yuyao
Pod Name: Baal (Group 1)

Working Memory

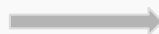
- Temporary storage and manipulation of information
- Critical for reasoning, decision-making, planning
- Neural correlate: stimulus-selective persistent activity

- Must be stable against noise and distraction
- Must be flexible to update/clear old contents
- Single-neuron firing can't be too rigid—or we'd never switch

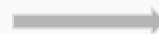
Short-Term Synaptic Plasticity:

Short-Term Facilitation (STF)

Transient INCREASE in strength when presynaptic firing is elevated



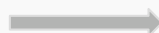
**“Deeper”
attractor basins**



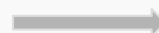
**Greater noise
resistance**

Short-Term Depression (STD)

Transient DECREASE in strength after high-rate firing



**“Shallow”
attractor basins**



**Easier
switching**

Where Attractor Networks Come In:

Attractor networks are recurrent circuits whose activity “settles” on fixed points.

They can be ...

Discrete : one stable pattern per learned memory

Continuous: a continuum of states

Energy landscape: “Basins” of attraction

STSP + Attractors

- STSP endows model with *adaptive* basin depth
- Synaptic dynamics tune working memory trade-off

Research Question



How does the balance between short-term facilitation and depression shape the functional trade-off between stability and flexibility in working memory?

What Do We Expect To Find?



Hypothesis 1

We first anticipate that **facilitation-dominant** networks will promote persistent, **stable** attractor states, supporting **maintenance of memory**.



Hypothesis 2

We anticipate that **depression-dominant** networks will promote faster updating, enabling **flexibility**, yet introduce **destabilisation** at the cost of memory longevity

Model Definitions

★ Memory



MEMORY
PROBLEMS

Persistent activity of a neural population (E1 or E2) after stimulus removal

★ Flexibility



Ability of the network to switch from one memory to another (switch to E2 after second stimulus)

★ Stability



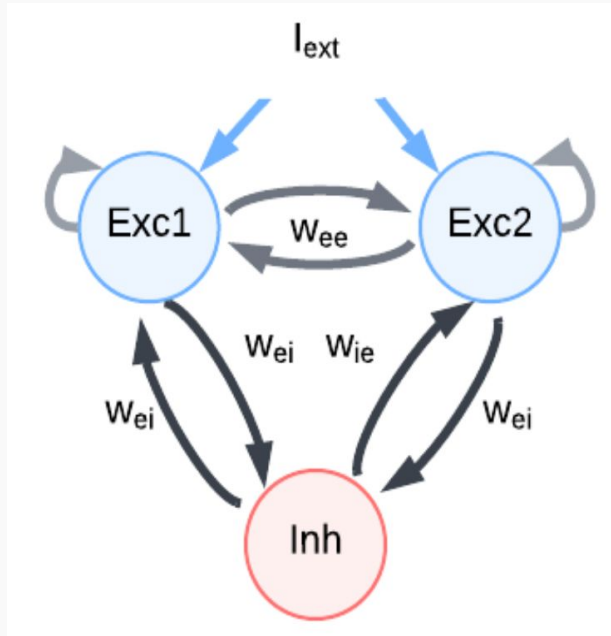
Resistance to change (hold onto E1 activity after stimulus ends)

★ Trade-off



Relationship stability vs. flexibility across combinations of STP parameters

Model Architecture



Exc1, Exc2: Excitatory neurons

Inh: Inhibitory neuron

W_{ee} : Excitatory to excitatory weight

W_{ei} : Excitatory to inhibitory weight

W_{ie} : Inhibitory to excitatory weight

I_{ext} : Stimuli from external input

- 1) External stimulus acts as an input that activates excitatory neurons
- 2) Excitatory loops established based on weights
- 3) Recurrent activity of the loops enable the memory state to be sustained over time
- 4) Excitation activates the inhibitory neuron which gives a feedback inhibition stabilizing the network

STF can be displayed by the model by temporarily strengthening the excitatory weights during activity representing stability in the maintenance of memory states.

In contrast, STD is displayed by the model by weakening the weights over time in order to allow the model the flexibility of transition between memory states.

Methodology - Simulating Working Memory

The **Wilson–Cowan model** describes the dynamics of interactions between homogenous populations of simple excitatory and inhibitory model neurons.

In our case, we have 2 excitatory units and 1 inhibitory unit interconnected by certain connection weights. The Wilson-Cowan model then helps us model the dynamics of the excitatory activities (and subsequently the inhibitory activities) and their firing rates wrt time t .

$$\begin{aligned}\frac{dE_1}{dt} &= -E_1 + \sigma(W_{EE}E_1 + W_{EE}E_2 - W_{IE}I + I_{ext,E1}; k, \theta) \\ \frac{dE_2}{dt} &= -E_2 + \sigma(W_{EE}E_1 + W_{EE}E_2 - W_{IE}I + I_{ext,E2}; k, \theta) \\ I_{instan} &= \sigma(W_{EI}E_1 + W_{EI}E_2; k, \theta)\end{aligned}$$

Where W_{EE} , W_{EI} and W_{IE} are the connection weights, inhibition is instantaneous, and I_{ext} are the external inputs on E_1 and E_2

Nullcline is a curve (or surface) in a system's state space where the rate of change of one variable is zero.

i.e. where $dE_1/dt = 0$ and $dE_2/dt = 0$

The interaction between these nullclines determines where and how the system and memory settles.

Experiments:

Visualizing nullclines and simulating trajectories of excitatory activities using the WC model to understand presence and stability of memory

Using a stable network configuration from nullcline analysis to simulate working memory

Methodology - Analyzing Stability and Flexibility with Plasticity

Tsodyks - Markram Dynamics

The Tsodyks-Markram (TM) model describes short-term synaptic plasticity (STSP), focusing on how synaptic connections between neurons change with repeated firing.

It's a mathematical model that incorporates both

- synaptic depression (reduced neurotransmitter release after repeated spikes) and,
- synaptic facilitation (increased release probability)

$$\frac{dx}{dt} = \frac{1-x}{\tau_{rec}-uxE}$$

$$\frac{du}{dt} = \frac{U-u}{\tau_{facil}+U(1-u)E}$$

where $x(t)$ is the available synaptic efficacy (related to STD)
 $u(t)$ is the release probability (related to STF)
and E is the presynaptic firing rate

Experimental Setup:

In a 2-excitatory and 1-inhibitory attractor system, stimulate E1 first for a few ms -> wait for E1 memory to stabilize and then stimulate E2.

For **memory stability**:

E1 memory stays up even after E2 is stimulated and no memory switch is observed.

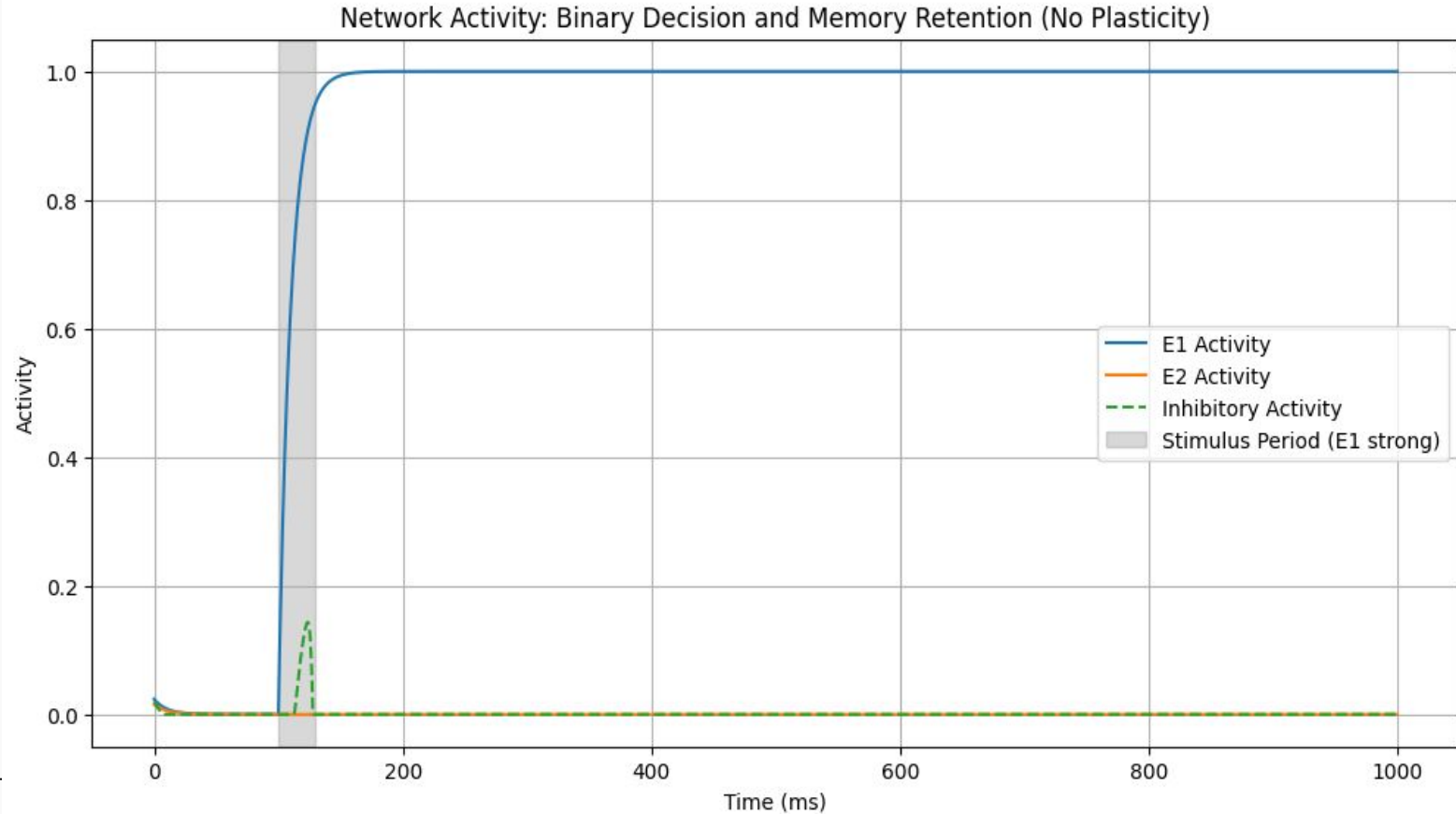
For **memory flexibility**:

E2 fires after it is stimulated and memory switches from E1 to E2.

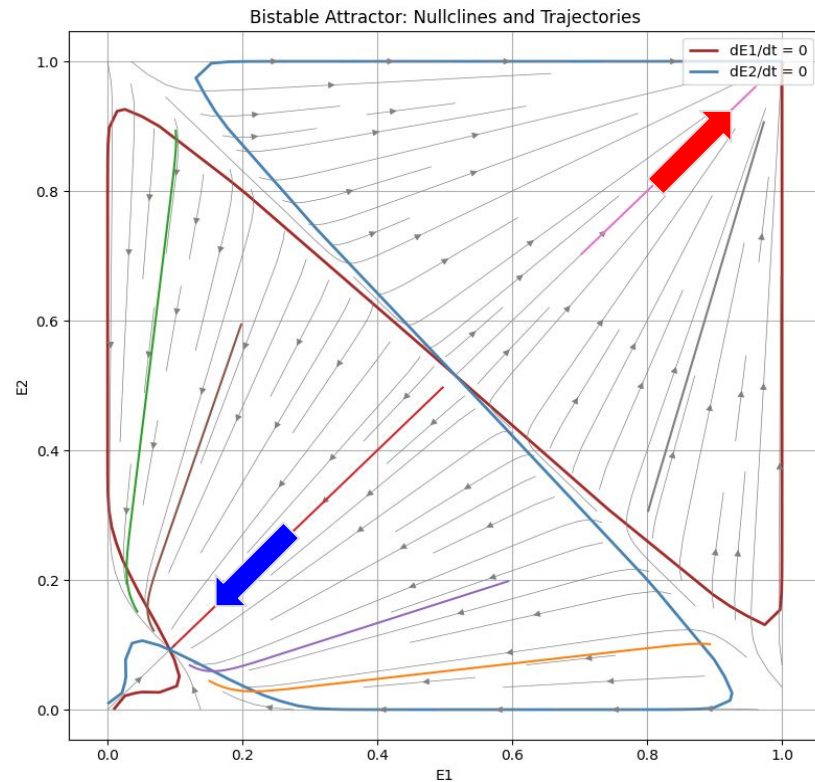
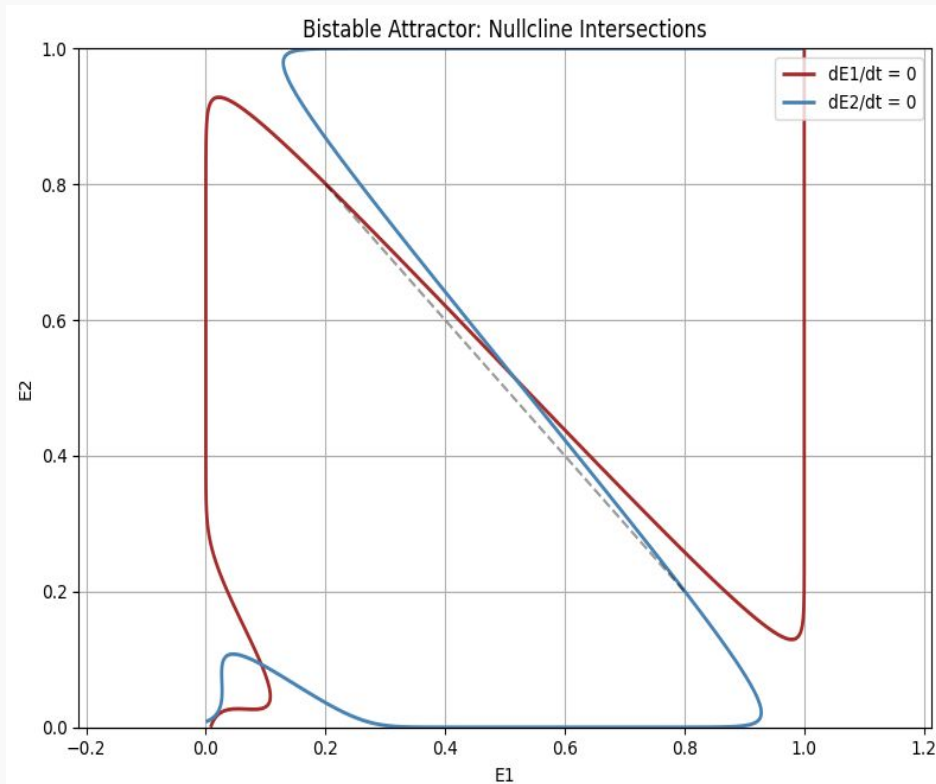
Hypothesis:

- $\text{STF} \propto \text{Stability} \propto 1/\text{Flexibility}$
- $\text{STD} \propto \text{Flexibility} \propto 1/\text{Stability}$

2E/1I Working Memory Simulation with E1 stimulus

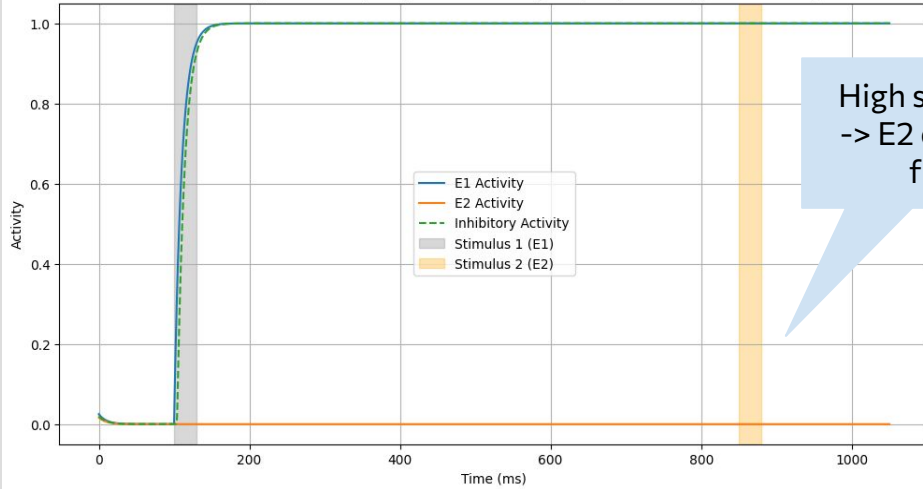


Nullclines Analysis

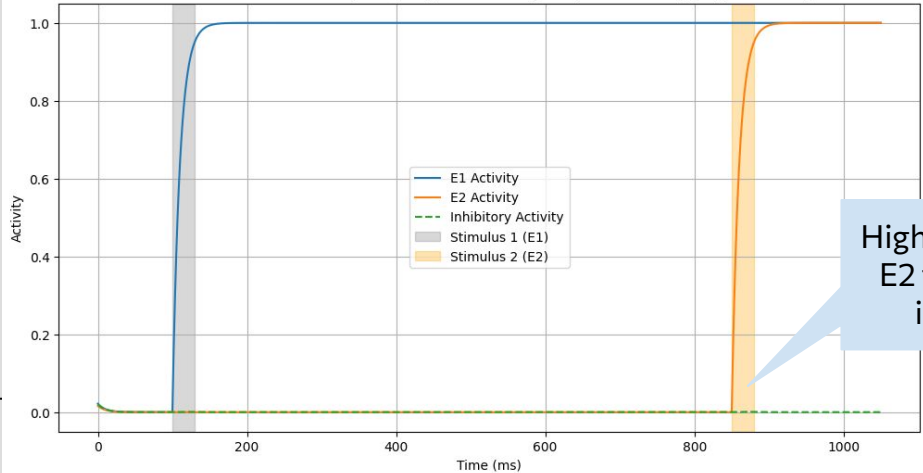


Analyzing Stability and Flexibility with Plasticity

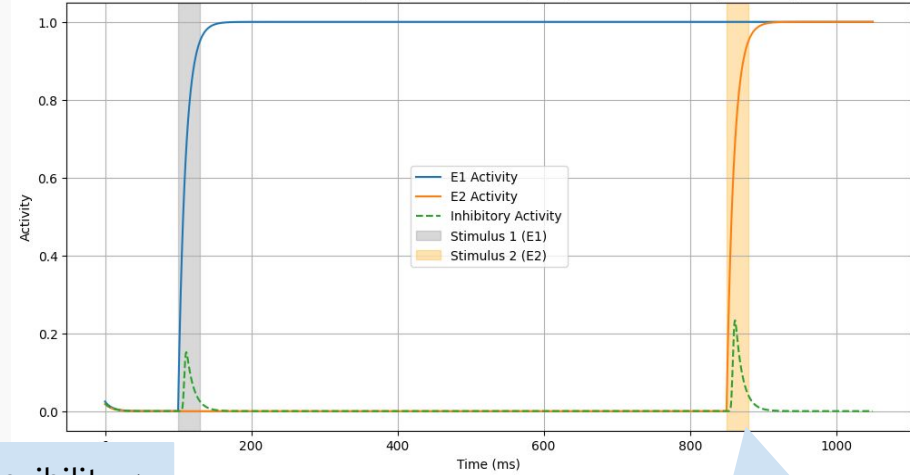
Network: High STF Scale (1.80), Low STD Scale (0.02) - High Stability, Low Flexibility



Network: Low STF Scale (0.02), High STD Scale (1.80) - Low Stability, High Flexibility



Network: Mid STF Scale (1.00), Mid STD Scale (1.00) - Balanced Stability/Flexibility



Why This Matters

Results

- Stable plateaus after brief input — modeling memory retention
- Switchable responses — modeling memory updating or loss

Theory

- Attractor dynamics: Multiple stable states (memory vs. rest)
- NMDA currents sustain activity — relevant in schizophrenia
- Short-term plasticity tunes memory stability vs. flexibility

Application

- Cognitive neuroscience: Explains working memory mechanisms
- Psychopathology: Models memory deficits in psychiatric disorders
- Computational psychiatry: Patient-specific simulations for interventions

Limitations & Future direction

- Validating the model against experimental working memory data to assess its biological relevance.
 - Expanding to larger or layered recurrent networks to test whether the stability-flexibility trade-off persists in high-dimensional systems.
 - Exploring how STSP influences the trade-off between stability and flexibility of working memory.
 - Temporarily modifying synapses' efficacy in response to stimuli
 - Influencing drift and diffusion of the system
-

Conclusion & Key Takeaways

Computational model

- 2E/1I attractor model with STSP
- Stimulates working memory with biologically realistic mechanisms
- Exhibits a bistable structure (E1-dominant vs. E2-dominant)

Stability & flexibility trade-off

High STF + Low STD
= **Higher Stability**

Low STF + High STD
= **Higher Flexibility**

Implications

- Shows how biological neural networks can implement persistent memory states
- Provides a foundation for future validation and potential translation to psychiatric disorders

Thank You!

We would like to acknowledge and thank the following Neuromatch contributors for their guidance and support:

Thank you to Peter Ohue, Sara Rostami, Dominik Endres, and Lorenzo Fontolan for their invaluable suggestions and guidance.
